

DecodeLabs Data Analytics - Project 1

Data Cleaning & Preparation Report

Prepared by: Anuj Kacher
Batch: 2026
Dataset: E-Commerce Sales Orders

Project Objective

The objective of this project is to clean a raw e-commerce transaction dataset by identifying missing values, removing duplicates, standardizing formats, validating numeric calculations, and preparing a reliable dataset for future analysis and dashboards.

Dataset Overview

Metric	Value
Total records before cleaning	1200
Total records after cleaning	1200
Duplicate full rows removed	0
Duplicate Order IDs found	0
Missing CouponCode values handled	309
Invalid dates after cleaning	0
TotalPrice mismatches found	0
Final missing/blank critical values	0

Columns Cleaned

Column	Cleaning/Validation Done
OrderID	Checked uniqueness and duplicate identifiers
Date	Converted to YYYY-MM-DD format
CustomerID	Trimmed and validated ID format
Product	Standardized product category names
Quantity, UnitPrice, TotalPrice	Converted to numeric format and checked calculation
ShippingAddress	Standardized address text
Payment/Status/Referral	Standardized categorical text
CouponCode	Blank values replaced with "No Coupon"
TrackingNumber	Trimmed and checked text format

Cleaning Methodology

Step 1 - Missing Values: Missing coupon values were replaced with "No Coupon" because a blank coupon means the customer did not use a promotion. Critical identifiers such as OrderID, CustomerID, and TrackingNumber were checked for blanks.

Step 2 - Duplicate Audit: Full-row duplicates and duplicate OrderID values were checked. This avoids inflated transaction counts and prevents revenue overstatement.

Step 3 - Format Standardization: Dates were converted to ISO format YYYY-MM-DD. Text columns were stripped of extra spaces. Numeric columns were converted to numbers and rounded to two decimals.

Step 4 - Calculation Validation: TotalPrice was checked against Quantity x UnitPrice. Mismatches were flagged in the cleaned dataset.

Verification Results

Check	Result
Duplicate Order IDs	0 found
Incorrect date formats	0 after conversion
Numeric conversion errors	0 in key numeric columns
TotalPrice validation	0 mismatches flagged
Ready for analysis	Yes

Exploratory Summary After Cleaning

Top Product Revenue

Product	Revenue
Chair	195620.11
Printer	195612.61
Laptop	192126.56
Tablet	186568.95
Monitor	175651.41
Desk	167459.93
Phone	151722.39

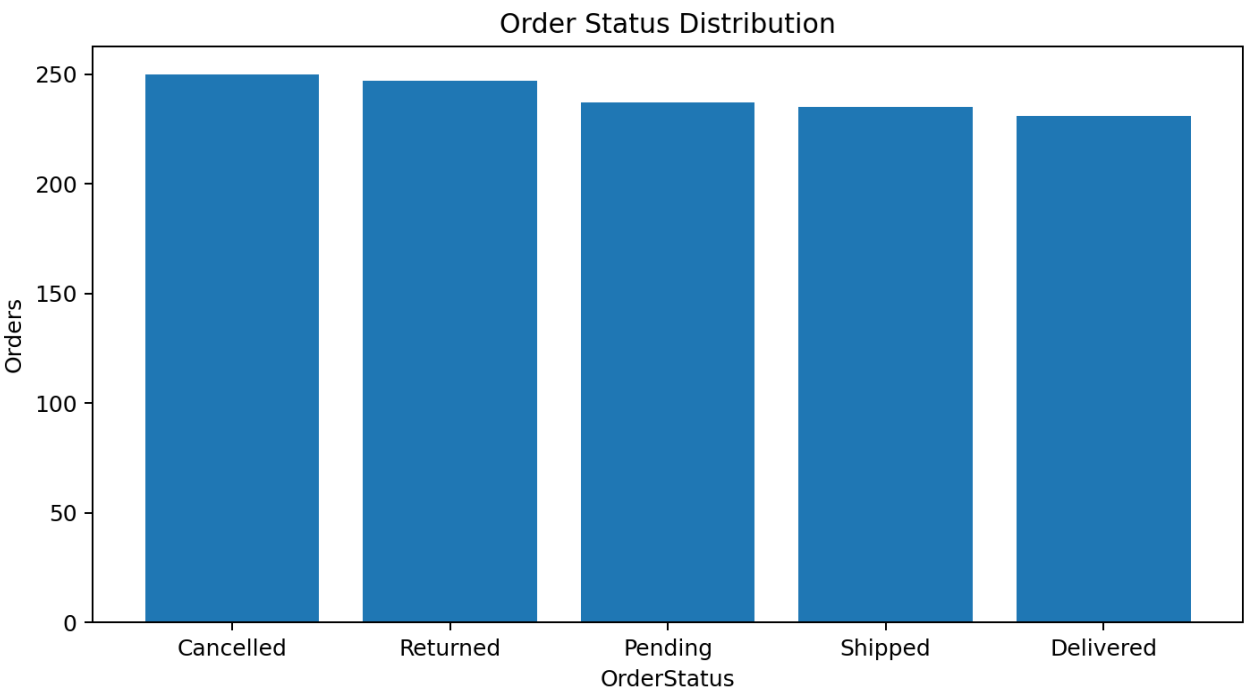
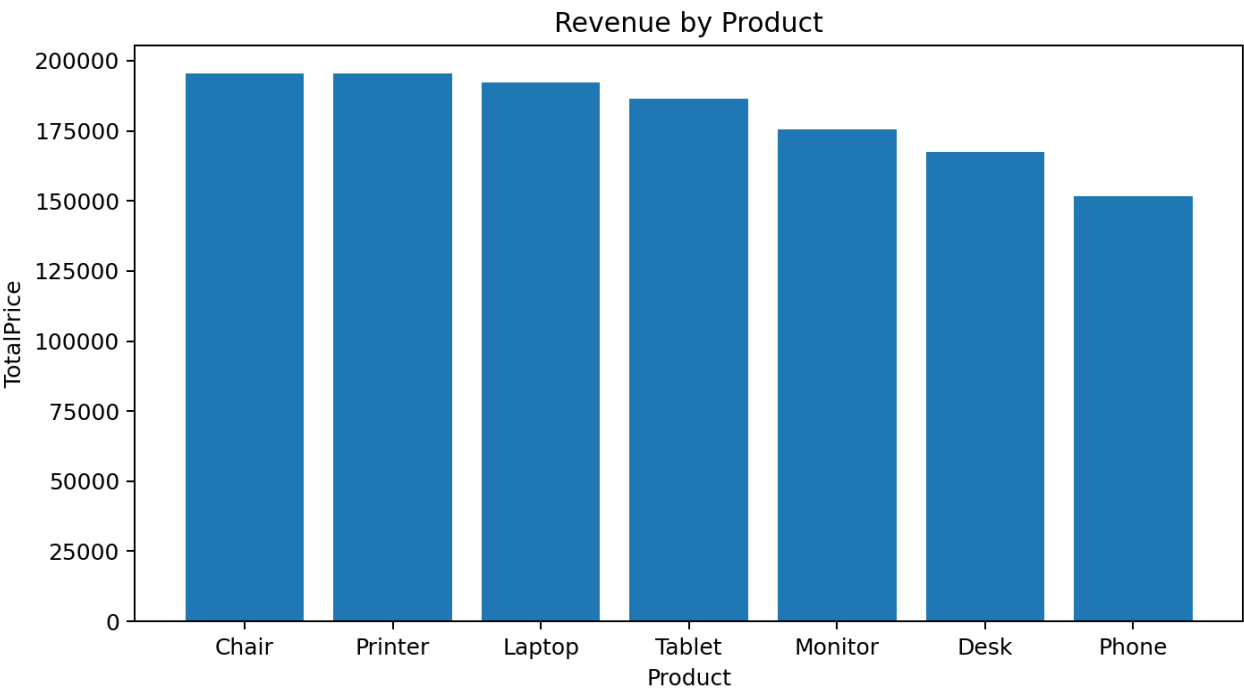
Order Status Distribution

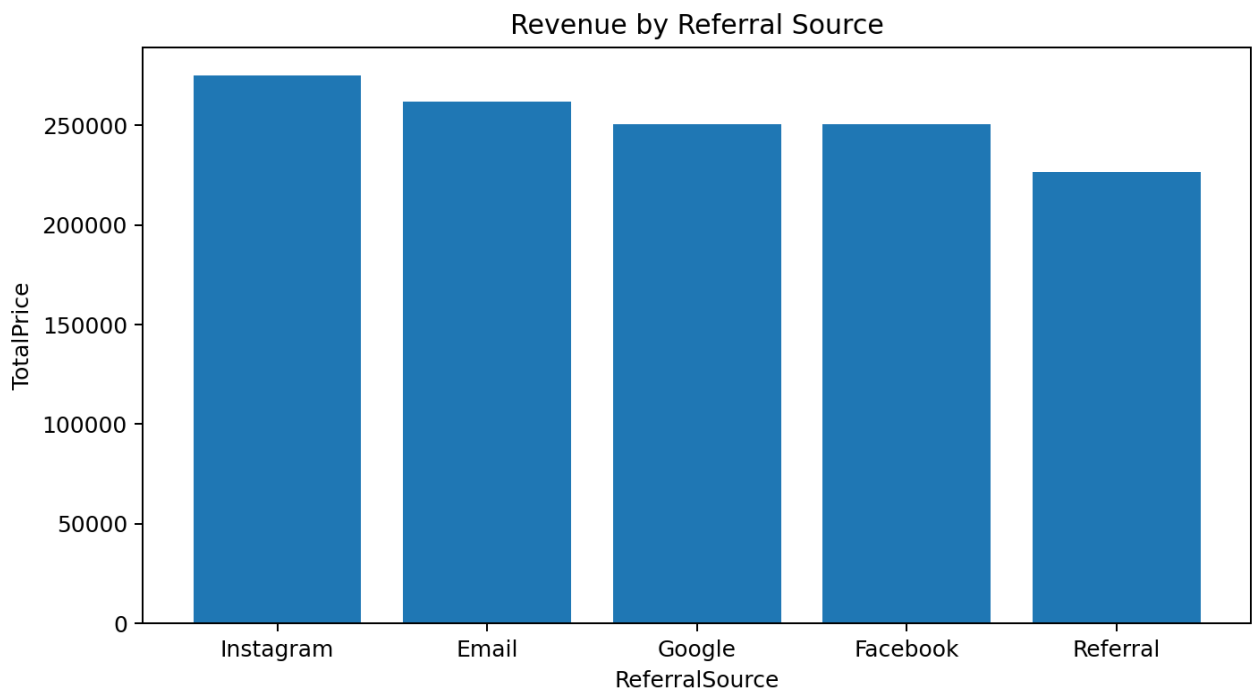
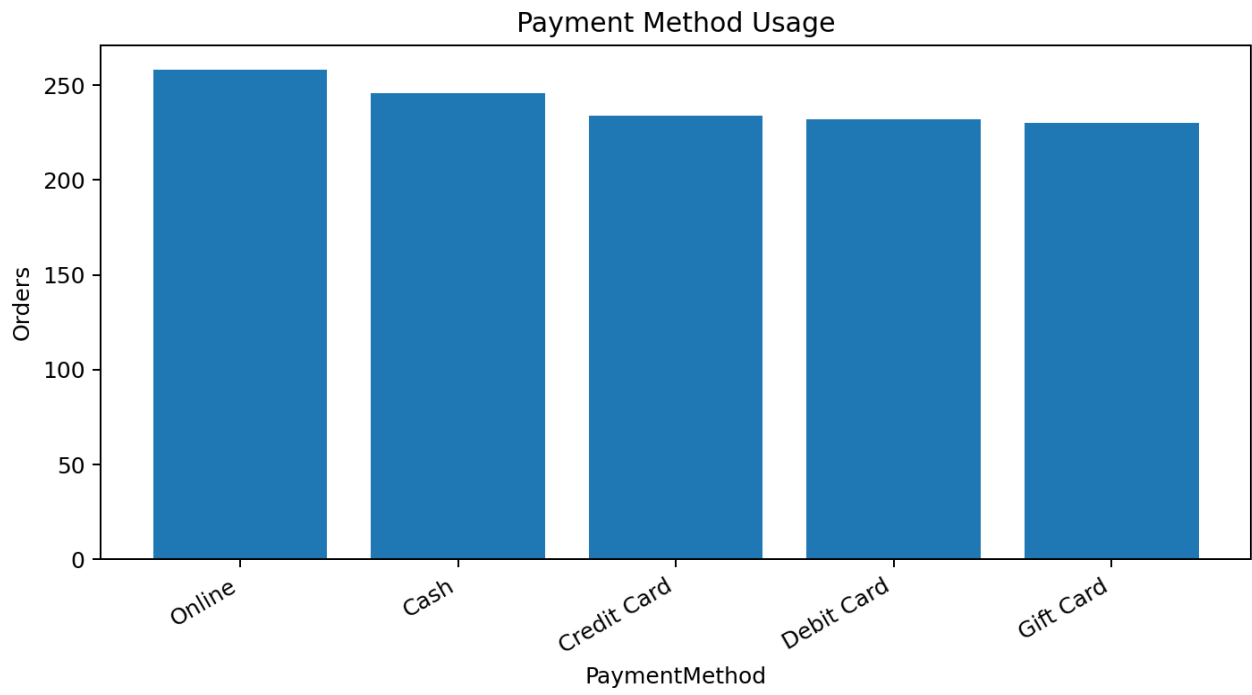
Order Status	Orders
Cancelled	250
Returned	247
Pending	237
Shipped	235
Delivered	231

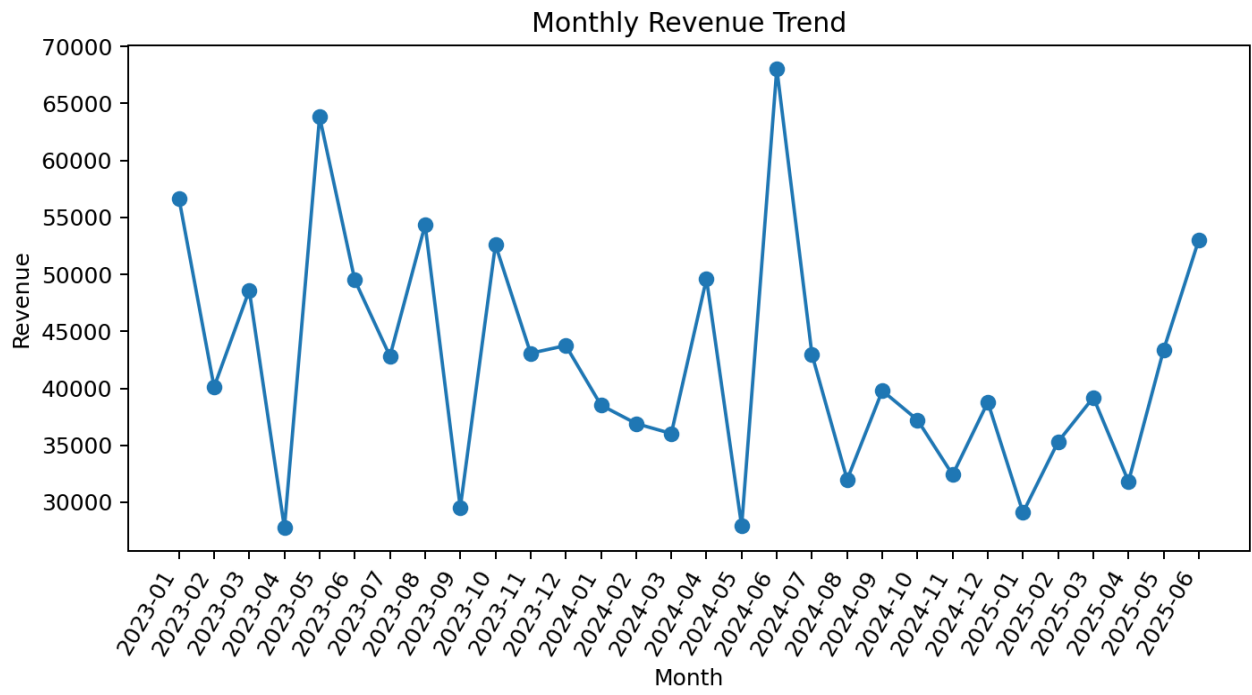
Revenue by Referral Source

Referral Source	Revenue
Instagram	275285.45
Email	261808.55
Google	250441.48
Facebook	250410.9
Referral	226815.58

Charts







Change Log

Change ID	Description	Impact	Status
CR001	Parsed raw PDF table into structured rows	Created analyzable dataset	Completed
CR002	Standardized date field to YYYY-MM-DD	Correct filtering and sorting	Completed
CR003	Converted Quantity, UnitPrice, ItemsInCart, TotalPrice to numeric fields	Allowed for mathematical operations	Completed
CR004	Replaced missing CouponCode with No Coupon	Preserved all records	Completed
CR005	Checked duplicate rows and duplicate OrderIDs	Reduced risk of inflated counts	Completed
CR006	Added CalculatedTotal and TotalPriceMismatch columns	Verified revenue accuracy	Completed

Conclusion

The dataset has been cleaned and prepared according to the DecodeLabs Project 1 requirements. It now contains unique identifiers, standardized dates, numeric columns suitable for analysis, and documented transformations. The final cleaned dataset can be used for SQL analysis, Excel pivot tables, Power BI dashboards, and future data analytics tasks.